Affine Response in the Isolated-Dummy FIFO Parity Game

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Abstract

Let a finite simple graph be played by opening each vertex into a queue and later closing the queue front. Closing a vertex toggles a binary score by its degree into the untouched set. We study the conjecture that adjoining one isolated dummy vertex lets either designated player force terminal score zero. The statement remains open.

The legal schedule is graph-independent. We associate to every complete schedule a vector in the binary edge space whose pairing with the adjacency vector is the terminal score. For a fixed opposing strategy, universal success is therefore equivalent to zero lying in the affine hull of the compatible terminal vectors. A score-erased public-policy formulation is proved equivalent to the original graph problem by finite-field separation. FIFO order further gives a triangular cut-space filtration and exact affine recursions for complete reply fans.

We prove or record the strongest current reductions: matching-plus-isolates boards are solved for either seat; close-first counterstrategies are excluded; every separated public policy has a rank-minimal selected open whose complete reply fan is unseparated but leaves a unit of parity debt in the old queue; and strategy-relative minimization reduces a counterstrategy to two ancestry patterns, a charged-close spike or a protected singleton. The unresolved step is a causal factor extension: prefix cancellations must be lifted through branch-dependent continuation cosets using strictly earlier defender siblings. Exact countermodels show why positionality, local fan identities, unlabelled state-graph cycles, and unrestricted pair descent do not provide that extension. Lean checks the game semantics and the stated subsidiary reductions, but neither assumes nor proves the conjecture. Exact minimax through eight real vertices is consistent with it and has no universal force.

1 Introduction

Queue layouts constrain graph edges by a FIFO discipline [2]. The game considered here instead places the vertices themselves in a FIFO queue. Play determines a family of proper activity intervals in the sense of interval graph theory [3]; graph edges only evaluate the resulting schedule.

Let R be a finite set of *real* vertices, let G be a simple graph on R, and adjoin an isolated vertex d. Two players alternate moves. A designated *defender* seeks terminal score $0 \in \mathbb{F}_2$, while the other player seeks score 1. Either physical seat may be designated the defender.

Conjecture 1.1 (isolated-dummy FIFO linking)**.** *For every finite simple graph G and isolated dummy d, the defender can force terminal score zero from the initial state, whichever physical seat is designated defender.*

This conjecture is stronger than the theorem needed for the Gold–Arf game construction. That application first chooses a public Witt frame, reducing the support graph to a matching plus isolated vertices. The matching case is proved independently; no application elsewhere in Ogdoad depends on Conjecture 1.1.

The purpose of this article is twofold. First, it gives a self-contained affine formulation of the game and the FIFO contraction calculus. Second, it states the precise remaining theorem after the strongest known local and strategy-relative reductions. Proof, formal verification, exact finite evidence, and open assertions are kept separate throughout.

2 The game and its universal edge moment

Put $V = R \sqcup \{d\}$ and extend G by declaring d isolated. A state is

$$s = (U, Q, k, t, g),$$

where $U \subseteq V$ is the untouched set, Q is a list of distinct open vertices, $k \in \{0, 1\}$ is a ko bit, t is the player to move, and $g \in \mathbb{F}_2$ is the score. Reachable states have $U \cap Q = \varnothing$.

Definition 2.1 (moves)**.** The legal transitions are as follows.

1. OPEN(x), for $x \in U$, removes x from U and appends it to Q. It sets $k = 1$ exactly when the old queue was empty.
2. CLOSE, when $Q = (f, Q_0)$ and $k = 0$, removes f and changes the score by
$$\chi_G(f, U) = \deg_G(f, U) \pmod 2.$$
3. If neither an open nor a close is legal, a forced pass clears k.

Every transition changes the player to move. The initial state is $(V, (), 0, t_0, 0)$, and a state is terminal when $U = \varnothing$ and $Q = ()$.

The rank $4|U| + 2|Q| + k$ strictly decreases at every transition. The game is therefore finite and determined by backward induction. On a reachable nonterminal state, ko is set only on a singleton queue created by opening onto an empty queue. A forced pass can occur only after U is empty, so no later close can change the score.

For a complete history h, let o_x and c_x be the opening and closing times of x, and put $I_x = [o_x, c_x]$. FIFO order makes the opening and closing orders identical, hence the intervals are pairwise nonnesting. Let

$$\mathcal{E}_R = \mathbb{F}_2^{\binom{R}{2}}$$

be the binary edge space with basis xy for unordered pairs $\{x, y\} \subseteq R$, and define

$$D(h) = \sum_{\{x,y\} \subseteq R} [I_x \cap I_y = \varnothing], xy.$$

Write $A_G \in \mathcal{E}_R$ for the adjacency vector of G.

Proposition 2.2 (interval accounting)**.** *For every complete history h,*

$$g(h) = \langle A_G, D(h) \rangle.$$

Proof. Fix an edge xy and suppose that x opens before y. This edge contributes to a close charge precisely when x closes while y remains untouched. By FIFO order this is equivalent to $c_x < o_y$, hence to $I_x \cap I_y = \varnothing$. Summing over the edges proves the formula. □

Thus the graph does not affect legal play; it is a linear functional applied to a universal schedule invariant.

3 Affine duality

Fix an attacker seat and a deterministic attacker strategy S in the graph-independent schedule tree. The strategy may depend on the complete history. Let $\mathcal{H}_S$ be the terminal histories compatible with S and arbitrary defender choices.

Theorem 3.1 (affine separation)**.** *The strategy S forces score one for some graph on R if and only if*

$$0 \notin \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\}.$$

Consequently, Conjecture 1.1 is equivalent, for both choices of attacker seat, to

$$0 \in \mathrm{Aff}_{\mathbb{F}_2}\{D(h) : h \in \mathcal{H}_S\} \qquad \textit{for every deterministic } S.$$

Proof. If S forces score one on G, Proposition 2.2 puts every $D(h)$ in the affine hyperplane $\langle A_G, z\rangle = 1$, which excludes zero. Conversely, write the affine hull as $a + W$. If it excludes zero, then $a \notin W$, so finite-dimensional separation over $\mathbb{F}_2$ gives a linear functional ℓ satisfying $\ell(W) = 0$ and $\ell(a) = 1$. Every functional on $\mathcal{E}_R$ is pairing with the adjacency vector of a simple graph on R. The strategy S forces score one on that graph. □

Corollary 3.2 (odd response flow)**.** *The affine condition holds for S if and only if there is an odd formal sum of compatible terminal histories whose disjointness vectors sum to zero.*

Proof. An affine combination over $\mathbb{F}_2$ is exactly a linear combination whose coefficient sum is one. □

At attacker nodes such a flow follows the child selected by S; at defender nodes it may split among an odd number of children. It is an algebraic certificate, not a randomized strategy.

There is also an exact score-erased formulation. A *public policy* remembers the schedule state and the attacker's selected moves but forgets the graph and score. Its leaf vectors are projected away from coordinates incident to d.

Theorem 3.3 (public-policy duality)**.** *The following statements are equivalent.*

1. *Conjecture 1.1 holds.*
2. *For either attacker seat, the projected affine response space of every initial public policy contains zero.*

Proof. An odd strategy for an isolated-dummy graph forgets its score coordinate to give a public policy separated by that graph's adjacency functional. Conversely, if the projected affine response space of a public policy omits zero, separation supplies a functional on the real edge space. Regard its coordinates as the adjacency vector of a simple graph and declare d isolated. The policy lifts, by backward recursion, to an odd strategy for that graph. The two constructions are inverse at the level of existence. □

Theorem 3.3 is useful because it shows that the affine problem is not merely a relaxation of the game. Proving projected affine zero for public policies is exactly the original theorem.

4 The FIFO contraction calculus

4.1 Live stars and complete reply fans

For a live real set L and $x \in L$, put

$$s_L(x) = \sum_{y \in L \setminus \{x\}} xy \in \mathcal{E}_R.$$

The dummy star is zero. Assign $s_L(x)$ to an open of x, measured just before the open, and assign zero to closes and passes.

Proposition 4.1 (live-star identity)**.** *For every complete history,*

$$D(h) = \sum_{x \text{ opened}} s_{L_x}(x),$$

where L_x is the real live set immediately before x opens.

Proof. Suppose x opens before y. The coordinate xy appears when x opens and appears again when y opens exactly if x remains live. The two copies then cancel. It appears once precisely when the activity intervals are disjoint. □

For a queue Q and untouched set U, write

$$P_G(Q, U) = e_G(Q, U) \pmod 2$$

for the scalar queue cut. Closing the front changes both P_G and the score by the same bit. Let P_x be the queue cut after opening $x \in U$, and P_{xy} the cut after subsequently opening $y \neq x$.

Proposition 4.2 (complete reply-fan identity)**.** *For every well-formed state,*

$$\sum_{y \in U \setminus \{x\}} P_{xy} = |U| P_x \qquad \text{in } \mathbb{F}_2.$$

If $|U|$ is even, the second-open replies form an odd family with zero aggregate cut. If $|U|$ is odd and the old queue is nonempty, adjoining the legal front close, translated by its score charge, again gives an odd family with zero aggregate cut.

Proof. If $\deg_L$ is degree in the live graph, then

$$P_{xy} = P_G(Q, U) + \deg_L(x) + \deg_L(y).$$

Handshaking gives $\sum_{u \in U} \deg_L(u) = P_G(Q, U)$. Summing the first identity over $y \neq x$ gives the formula. The parity consequences follow by counting the replies; in the odd case the translated close contributes the missing copy of P_x. □

The only nonterminal local exception is an odd untouched set with empty queue: the first open sets ko and there is no close reply. More importantly, Proposition 4.2 cancels only prefix cuts. Below its children the attacker chooses different continuation cosets.

4.2 Continuation cosets

At a node q of a fixed attacker strategy, let Λ_q be the compatible terminal leaves. For $\epsilon \in \mathbb{F}_2$, define

$$\mathcal{C}_q^\epsilon = \left\{ \sum_{\lambda \in \Lambda_q} e_\lambda D_q(\lambda) : \sum_\lambda e_\lambda = \epsilon \right\}.$$

Then $W_q = \mathcal{C}_q^0$ is a vector space and $A_q = \mathcal{C}_q^1$ is a nonempty affine W_q-coset.

Proposition 4.3 (affine Bellman recursion)**.** *Suppose a defender node q has children q_m with move vectors c_m. Choose $a_m \in A_{q_m}$ and one child m_0. Then*

$$W_q = \sum_m W_{q_m} + \operatorname{span}\{(c_m + a_m) + (c_{m_0} + a_{m_0}) : m \neq m_0\},$$
$$A_q = c_{m_0} + a_{m_0} + W_q.$$

At an attacker node with selected child q' and move vector c, $W_q = W_{q'}$ and $A_q = c + A_{q'}$.

Proof. A leaf family splits uniquely among the child leaf sets. Internal even families give $\sum_m W_{q_m}$. Relative to the odd profile supported at m_0, every other odd child-parity profile is obtained by toggling m_0 with another child. These toggles give the displayed generators. The attacker formula is immediate because only one child remains. $\square$

If an antichain $\mathcal{F}$ meets every terminal branch below a root once, and Γ_q is the root-to-q prefix moment, iteration gives

$$A_{\text{root}} = \operatorname{Aff}\{\Gamma_q + a_q : q \in \mathcal{F}\} + \sum_{q \in \mathcal{F}} W_q.$$

This is the exact interface between local prefix cancellation and correlated future play.

4.3 Front deletion

For a current live set L and real queue front $f \in L$, put

$$\operatorname{Cut}(L) = \operatorname{span}\{s_L(x) : x \in L\}.$$

Theorem 4.4 (front splitting)**.** *There is a canonical direct sum*

$$\mathcal{E}_L = \operatorname{Cut}(L) \oplus \mathcal{E}_{L \setminus \{f\}}.$$

The projection to the second summand is

$$(T_f z)_{ij} = z_{ij} + z_{fi} + z_{fj}, \qquad i, j \neq f.$$

It kills every live star and fixes vectors supported away from f.

Proof. The formula kills $s_L(f)$ and, for $x \neq f$, cancels the ij and fi coordinates of $s_L(x)$. It plainly fixes $\mathcal{E}_{L \setminus \{f\}}$. The only relation among the live stars is $\sum_{x \in L} s_L(x) = 0$, so $\dim \operatorname{Cut}(L) = |L| - 1$. The dimensions add to $\binom{|L|}{2}$, proving the direct sum. $\square$

The operator T_f is the f-chart of the usual switching/coboundary correspondence for two-graphs [4]. While f remains at the front, every open contributes a cut vector; after f closes, all later contributions lie in the smaller edge space. FIFO order therefore gives a triangular filtration. It does not let one choose continuation representatives independently of the prefixes: doing so would replace the correlated affine set in the preceding frontier formula by a larger Cartesian product.

5 Proved regimes

5.1 Paired boards

Theorem 5.1 (parity-cell pairing)**.** *Suppose V has a partition $\mathcal{M}$ into two-element cells satisfying*

$$e_G(A,B) = 0 \pmod 2 \qquad (A, B \in \mathcal{M}, A \neq B).$$

Then the second player can force terminal score zero.

Proof. After the first player opens one member of a cell, open its mate; after the first player closes the front member of a cell, close the next front. The untouched set remains a union of cells and the queue a concatenation of cells. Ko causes no exception because an open onto an empty queue is answered by the mate open. If $A = \{a, a'\}$ closes while the untouched set is a union of cells, its total charge is

$$\deg_U(a) + \deg_U(a') = \sum_{B \subseteq U} e_G(A,B) = 0.$$

Every close occurs in one such pair. □

There is a separate constructive theorem for matching-plus-isolates boards: either physical seat, when designated defender, can force score zero. Its safe-front induction is the exact theorem used by the Gold–Arf realization. Theorem 5.1 is neither needed to strengthen that application nor sufficient for arbitrary graphs.

5.2 Close-first strategies

Call an attacker *close-first* when it closes whenever the queue is nonempty and ko is clear, and opens only when a close is unavailable.

Theorem 5.2 (close-first exclusion)**.** *The following statements hold for every finite graph.*

1. *From a clear state with nonempty queue and the defender to move, the defender can force future score increment zero against a close-first attacker.*
2. *From a score-zero empty-queue root with the defender to move, the same remains true if the attacker may stop and claim a win at a clear score-one state.*

No dummy or cardinality hypothesis is required.

Proof outline. For the first assertion, choose a counterexample of minimum remaining rank and write the front as f. Opening each $x \in U$ before the forced close shows that every punctured charge $\deg_{U\setminus\{x\}}(f)$ must equal one. Hence f is adjacent to every member of U, $|U|$ is even, and $\deg_U(f) = 0$. Comparing two-open prefixes at the next front makes its row constant on U and neutral as well. Closing the neutral fronts produces a smaller counterexample, a contradiction. The cases $|U| \leq 1$ and a queue of length one are the direct endgames of the same calculation.

For the stopped empty-root statement, call an ordered pair $x \to y$ bad when the stopped attacker wins from the queue (x, y). Three zero-charge absorbers and the handshaking identities attach to every vertex a rank in $\{0, 1, 2\}$ which strictly decreases along each bad arc. If the root were bad, every possible first open would have an outgoing bad reply, so the finite bad-arc graph would contain a directed cycle. Strict rank descent forbids such a cycle. Strong induction on the remaining carrier closes the absorber argument. □

Thus every hypothetical counterstrategy at the isolated-dummy root contains a node at which it opens although a close is legal. This fact is global only after the actual strategy ancestry is retained.

6 Normal form of a hypothetical counterstrategy

Two complementary minima describe the current frontier. One is dual and graph-free; the other is taken inside a fixed odd strategy on its separating graph.

6.1 The public separator minimum

Suppose an initial public policy omits projected zero. Choose a separating functional ℓ, and evaluate every live-star prefix by ℓ. A policy occurrence lies on *sheet* $b \in \mathbb{F}_2$ when its prefix value plus the constant affine value of its continuation coset is b.

Theorem 6.1 (separator normal form)**.** *In a constantly-one initial public policy, choose a sheet-one occurrence of minimum remaining rank while the dummy is live. Then:*

1. *it is an attacker occurrence and its selected move is a real open* $\mathrm{OPEN}(v)$ *of separator increment one;*
2. *the selected child and every edge in its complete defender reply fan lie on sheet zero and have separator increment zero;*
3. *the old queue Q is nonempty and*

$$\sum_{q\in Q} \ell\big(\pi_d s_L(q)\big) = 1,$$

   where L is the current live real set and π_d deletes dummy-incident coordinates;

4. *the resulting legal close sibling enters a selected sheet-zero continuation, but need not select another close.*

Proof outline. Rank minimality excludes a sheet-one selected child. A selected close or pass has zero live-star increment, so the selected move must be a real open of increment one. Every defender child must be present in the policy; a sheet-one child would contradict minimality, giving the complete zero sheet. The dummy open is among these zero-increment children while the dummy is live. Summing the live-star evaluations over the complete open fan and using handshaking moves the odd partner of v into the already open queue, giving the displayed debt relation. In particular Q is nonempty and close is legal. Sheet recursion gives the last statement, but imposes no action-type constraint on the next selected move. □

The theorem locates the missing parity partner exactly: it is FIFO debt, not another legal open sibling.

6.2 The strategy-relative minimum

On a fixed separating graph, minimize a score-zero failure inside one exact odd strategy while retaining its root prefix and its complete immediate defender fan.

Theorem 6.2 (minimal-bad ancestry)**.** *Every such minimum has one of two predecessor forms:*

1. *a unit-charged close followed by a score-neutral selected suffix (the* charged-close spike*);*
2. *an open onto an empty queue, producing a ko-protected singleton.*

A neutral-close predecessor and a clear-open predecessor are impossible.

Proof outline. Score translation identifies the odd continuation below a unit-charged close with a neutral continuation on the opposite sheet. Choosing a rank-minimal score-zero occurrence forces every proper selected descendant on that sheet to be pointwise score-neutral. A neutral close would therefore expose a strictly smaller failure. For a clear open, combine the close sibling with the complete family of open siblings. The queue-cut and live-star fan identities cancel the common prefix, while the fixed-front contraction eliminates the remaining local leaf. This again gives a smaller failure. The only open not covered by that fan is the ko-protected opening onto an empty queue, and the only surviving close is unit charged. □

Every score-one root prefix also has a last unit-charged close followed by a neutral suffix. The queued separator-debt vertex from Theorem 6.1 can either survive behind that front or be opened inside the suffix. Ko legality and score parity do not identify it with the charging neighbor in either order. Relating the two minima therefore requires their common initial-root ancestry.

7 The causal factor extension

For a node q in the fixed attacker strategy, retain the continuation coset $A_q = a_q + W_q$ of Section 4. Let Γ_q be the root-to-q prefix moment, projected away from dummy coordinates.

Proposition 7.1 (ancestry factor interface)**.** *Let $q_1, \dots, q_{2r+1}$ be occurrences in the same strategy tree. Choose $a_i \in A_{q_i}$ and $d_i \in W_{q_i}$. If*

$$\sum_i (\Gamma_{q_i} + a_i) = \sum_i d_i,$$

then the projected affine response space at the root contains zero.

Proof. Because d_i is an even continuation direction, $a_i + d_i \in A_{q_i}$. Lifting each point through its actual ancestry gives $\Gamma_{q_i} + a_i + d_i$ in the root affine response space. An odd sum of points in one affine space remains in that space, and the displayed factor identity makes the sum zero. □

Without restrictions on the q_i, Proposition 7.1 is tautological: one may choose the root. Its content is to build the odd family causally from the complete defender fans exposed by Theorems 6.1 and 6.2.

Conjecture 7.2 (causal factor extension)**.** *For every separated initial public policy, the charged-close-spike or protected-singleton frontier admits an odd family of occurrences satisfying that factor identity, using descendant continuation directions together with strictly earlier defender siblings.*

Theorem 7.3 (reduction to causal factor extension)**.** *Conjecture 7.2 implies Conjecture 1.1.*

Proof. If FIFO linking failed, Theorem 3.3 would give a separated initial public policy. Theorems 6.1 and 6.2 reduce it to one of the two stated frontiers. Conjecture 7.2 and Proposition 7.1 would put zero in its root response space, contradicting separation. □

Equivalently, if h is the selected minimal-bad occurrence, $b_h = \Gamma_h + a_h$, and M_h is generated by frontier continuation spaces and even sums of off-spine decorated points, the remaining membership is

$$b_h \in M_h.$$

The separating graph pairs to one with b_h and annihilates M_h in a hypothetical counterexample. The issue is therefore not another scalar parity identity; it is compatible selection across correlated continuations.

7.1 Sharp limits of local globalization

The following statements are proved by explicit finite games or exact affine models. They delimit, but do not refute, Conjecture 7.2.

1. Every odd-winning state has a canonical full-state positional strategy. Nevertheless, equal public states may lie on opposite score sheets, and cycles in the quotient state graph can carry a nonzero edge label.
2. The history-occurrence tree is acyclic and has injective edge boundary. Hence an unlabelled tree or state-DAG cycle supplies no zero moment by itself.
3. A neutral isolated-dummy open/close square transports schedules exactly, but deleting the dummy reverses the controller of the intervening real events. Root outcome values do not contain enough contextual data to splice the two sides.
4. A complete zero-prefix fan transports the affine unit into its continuation residue. Immediate fan completeness does not force an earlier sibling to select the lag-removing close.
5. Unrestricted equal-degree, zero-second-moment pair descent is false on $K_{1,4}$. This graph has no isolated vertex; adjoining an isolated dummy repairs the displayed pair, but no dummy-conditioned induction is known.
6. No uniformly bounded number of terminal response vectors can witness every required affine zero: causal simplex examples have unbounded minimal odd support.

Each item removes a tempting loss of information. None constructs an arbitrary-order counterexample to FIFO linking.

8 Formal verification and finite evidence

The formal development uses Lean 4 [1] and Mathlib [6]. The import-only paper entry point gathers the unified `FifoFrontier` surface. Its mathematical boundary is summarized in Table 1.

Formal group	Kernel-checked content
Game semantics	Legal transitions, rank descent, determinacy, score translation, terminal and singleton tails, and edgeless boards.
Affine/public duality	Strategy-indexed affine spaces, public-policy response spaces, separator reconstruction by isolated-dummy graphs, and the equivalence in Theorem 3.3.
Local contraction	Live-star and queue-cut identities, complete reply fans, front normalization, matching-plus-isolates, and close-first exclusion.
Counterstrategy normal form	Public separator minimality, queued debt, canonical positional strategies, last-charged-close decomposition, and the two surviving cases of Theorem 6.2.
Boundary models	The exact failures of unlabelled state-DAG incidence, unrestricted pair descent, local third-sibling selection, score-only block splicing, and dummy deletion.

Table 1: Lean-checked content relevant to the FIFO frontier.

The proposition named `FifoLinkingTheorem` in the formal development is a definition of the open target. It is neither an axiom nor a theorem. No collection of the subsidiary modules is presented as a kernel-checked proof of Conjecture 1.1. The formal project contains no `sorry`, `admit`, or custom axiom for this problem.

The executable model performs exact minimax on the same transition system. With one isolated dummy adjoined, every nonisomorphic graph through eight real vertices is even-winning for either designated seat. At eight real vertices the exhaustive layer contains 12,346 graph classes, enumerated with the nauty toolchain [5]. This is an exact finite census, not a proof for arbitrary order.

One exact nine-real-vertex example shows the sharpness of the first-seat selector. On real vertices $0, \ldots, 8$ take

$$\begin{aligned} E = \{&01, 02, 03, 04, 05, 06, 07, 08, 13, 15, 25, 28, 34, 35, 36, 37, 38, \\ &46, 57, 67, 68, 78\}. \end{aligned}$$

After the attacker opens 0, every real reply loses under exact minimax; only the dummy reply wins. That winning child also has the direct parity-cell certificate

$$\{0, d\}, \quad \{1, 2\}, \quad \{3, 8\}, \quad \{4, 5\}, \quad \{6, 7\}.$$

Thus even the first local selector can depend essentially on the dummy. The example is not an arbitrary-order obstruction.

9 Conclusion

The isolated-dummy FIFO linking problem is exactly an affine response problem in the real binary edge space. Interval disjointness, live-star sums, queue-cut parity, and front deletion are compatible descriptions of the same invariant. They solve paired boards, exclude close-first strategies, and reduce any hypothetical counterstrategy to a precise ancestry-sensitive normal form.

The remaining theorem is the causal factor extension. A rank-minimal public separator leaves its parity partner as debt in the old queue, while a strategy-relative minimum leaves either a charged-close spike or a protected singleton. The task is to select an odd family of strictly earlier defender siblings whose decorated prefixes and continuation directions cancel that debt. Until such a factor is proved, or an arbitrary-order separated policy is constructed, Conjecture 1.1 remains open.

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